Depth Estimation with Depth Anything V2 on Video

For this task, the freiburg1_xyz sequence from the TUM RGB-D dataset[1] was used as the input video. The video contains 798 frames with a resolution of 640 × 480 pixels and an input frame rate of 30 FPS. The total duration of the video is approximately 26.6 seconds.

The video processing pipeline was implemented as follows:

Input Video → Read Frame → Depth Anything V2 → Depth Map → Output Video

Each frame was read sequentially using OpenCV. The frames were converted from BGR to RGB and passed to **Depth Anything V2 Small**. The predicted depth map was then resized to the original frame resolution and converted into a color visualization. The resulting depth maps were written frame by frame to an output video.

Performance Results

Metric	Result
Input FPS	30 FPS
Number of frames	798
Resolution	640 × 480
Video duration	26.6 s
Model	Depth Anything V2 Small
Average inference time	60.01 ms/frame
Inference FPS	16.66 FPS
Total inference time	47.89 s
End-to-end processing time	60.33 s
End-to-end FPS	13.23 FPS

The **inference FPS of 16.66** represents the processing speed of the Depth Anything model itself. The **end-to-end FPS of 13.23** includes the complete pipeline, including frame reading, preprocessing, inference, depth resizing, and depth visualization.

Since the input video is 30 FPS, maintaining real-time processing at the original frame rate would require processing each frame within approximately 33.3 ms. The model requires approximately 60.01 ms per frame for inference. Therefore, the system **cannot** process every incoming frame in real time at 30 FPS on the current hardware.

The generated output video was successfully produced and visually inspected. The depth maps were correctly generated for all 798 frames.

[1] The TUM RGB-D dataset is a widely used benchmark for RGB-D computer vision research. It provides synchronized RGB and depth data captured with an RGB-D camera in indoor environments. https://cvg.cit.tum.de/data/datasets/rgbd-dataset/download